

Market Liquidity Model Feasibility Study

Integration of Liquidity Risk into Value at Risk Framework

Author: F.Balloni
Technical Documentation

May 7, 2026

Abstract

This feasibility study demonstrates the integration of market liquidity risk into a comprehensive Value at Risk (VaR) framework. We present three core objectives: (1) development of a liquidity buffer adjustment for VaR calculations, (2) validation of the 30-day VaR holding period using the Almgren-Chriss optimal execution model, and (3) establishment of quantitative metrics for IFRS 13 fair value hierarchy classification. The framework covers equities, bonds, ETFs, and FX spot markets, incorporating both theoretical foundations and practical implementation strategies. Using synthetic portfolio examples, we demonstrate the feasibility and effectiveness of each component, providing an actionable roadmap for implementation within existing risk management systems. The study concludes that all three objectives are technically feasible with available internal data and computational resources, subject to careful parameter calibration and ongoing model validation.

Contents

1	Executive Summary	5
1.1	Motivation and Context	5
1.2	Key Findings	5
1.3	Recommendations	5
2	Liquidity Buffer for Value at Risk	6
2.1	Theoretical Framework	6
2.1.1	Standard VaR Formulation	6
2.1.2	Liquidity-Adjusted VaR (LVaR)	6
2.2	Liquidation Cost Model	6
2.2.1	Position-Level Liquidation Cost	6
2.2.2	Portfolio-Level Liquidation Cost	7
2.3	Liquidity Buffer Calculation	7
2.4	Asset Class Specifications	7
2.4.1	Equities	7
2.4.2	Bonds	7
2.4.3	ETFs	8
2.4.4	FX Spot	8
2.5	Worked Example: Multi-Asset Portfolio	8
3	Holding Period Validation via Almgren-Chriss	9
3.1	Optimal Execution Theory	9
3.1.1	Problem Formulation	9
3.1.2	Market Impact Model	9
3.2	Optimal Liquidation Strategy	9

3.2.1	Cost and Risk Decomposition	9
3.2.2	Objective Function	10
3.2.3	Analytical Solution	10
3.3	Optimal Liquidation Time	10
3.3.1	Time Horizon Selection	10
3.3.2	Holding Period Validation Formula	10
3.4	Asset-Specific Implementation	11
3.4.1	Equities	11
3.4.2	Bonds	11
3.4.3	ETFs	11
3.4.4	FX Spot	11
3.5	Worked Examples	12
3.5.1	Example 1: Liquid Large-Cap Equity	12
3.5.2	Example 2: Illiquid Small-Cap Equity	12
3.5.3	Example 3: Corporate Bond	13
3.6	Parameter Calibration	13
3.6.1	Temporary Impact (η)	13
3.6.2	Permanent Impact (γ)	13
3.6.3	Risk Aversion (λ)	13
3.7	Holding Period Validation Conclusion	14
4	IFRS Fair Value Hierarchy Classification	14
4.1	IFRS 13 Framework	14
4.1.1	Fair Value Hierarchy Definitions	14
4.1.2	Active Market Definition	14
4.2	Quantitative Classification Metrics	15
4.2.1	Composite Liquidity Score	15
4.2.2	Classification Thresholds	15
4.3	Asset Class Classification Rules	15
4.3.1	Equities	15
4.3.2	Bonds	16
4.3.3	ETFs	16
4.3.4	FX Spot	17
4.4	Implementation Algorithm	17
4.5	Worked Examples	17
4.5.1	Example 1: S&P 500 Large-Cap Stock	17
4.5.2	Example 2: Investment-Grade Corporate Bond	18
4.5.3	Example 3: Illiquid Microcap Stock	19
4.5.4	Example 4: Emerging Market FX	19
4.6	Governance and Monitoring	20
4.6.1	Classification Review	20
4.6.2	Edge Cases	20
4.7	Regulatory Alignment	20
4.7.1	SEC Guidance (ASC 820)	20
4.7.2	ESMA Guidelines	20
5	Integrated Framework	20
5.1	End-to-End Workflow	20
5.2	Module Architecture	20
5.2.1	Proposed Structure	20
5.3	Data Requirements	21
5.3.1	Input Data Specifications	21

5.4	Computational Complexity	22
5.4.1	Performance Estimates	22
5.5	Comprehensive Example: 50-Position Portfolio	23
5.5.1	Portfolio Composition	23
5.5.2	Results Summary	23
6	Feasibility Assessment	24
6.1	Data Infrastructure	24
6.1.1	Available Data	24
6.1.2	Data Quality Considerations	24
6.2	Technical Implementation	24
6.2.1	Technology Stack	24
6.2.2	Development Effort	25
6.3	Model Risk and Limitations	25
6.3.1	Key Risks	25
6.3.2	Model Validation	26
6.4	Integration with Existing Systems	26
6.4.1	VaR System Integration	26
6.4.2	Front Office / Trading	26
6.5	Regulatory and Audit Considerations	27
6.5.1	Documentation Requirements	27
6.5.2	Model Governance	27
6.6	Feasibility Conclusion	27
7	Implementation Roadmap	28
7.1	Phased Approach	28
7.1.1	Phase 1: Foundation (Weeks 1-4)	28
7.1.2	Phase 2: Almgren-Chriss Development (Weeks 5-9)	28
7.1.3	Phase 3: VaR Integration (Weeks 10-13)	29
7.1.4	Phase 4: IFRS Classification (Weeks 14-16)	29
7.1.5	Phase 5: Validation & Testing (Weeks 17-21)	30
7.1.6	Phase 6: Production Deployment (Weeks 22-24)	30
7.2	Resource Requirements	31
7.2.1	Team Structure	31
7.3	Risk Mitigation	31
7.3.1	Common Pitfalls and Mitigations	31
8	Regulatory Compliance	32
8.1	Basel III / Basel IV	32
8.1.1	Liquidity Coverage Ratio (LCR)	32
8.1.2	Net Stable Funding Ratio (NSFR)	32
8.1.3	Market Risk Capital (FRTB)	32
8.2	IFRS 13 Compliance	33
8.2.1	Fair Value Disclosure Requirements	33
8.2.2	Valuation Techniques	33
8.3	Internal Risk Framework Alignment	33
8.3.1	Risk Appetite Statement	33
8.3.2	Stress Testing	34
8.3.3	Internal Limits Framework	34
8.4	Regulatory Reporting	34
8.4.1	Required Reports	34
8.4.2	Audit Trail	34

9	Conclusions and Recommendations	35
9.1	Summary of Findings	35
9.1.1	Liquidity Buffer for VaR	35
9.1.2	Holding Period Validation	35
9.1.3	IFRS Classification	35
9.2	Critical Success Factors	35
9.3	Recommendations	36
9.3.1	Immediate Actions (Next 3 Months)	36
9.3.2	Medium-Term (6-12 Months)	36
9.3.3	Long-Term (12-24 Months)	36
9.4	Limitations and Caveats	36
9.5	Final Assessment	37
A	Python Code Snippets	37
A.1	Liquidity Buffer Calculation	37
A.2	Almgren-Chriss Optimal Execution	38
A.3	IFRS Classification Algorithm	41
B	Parameter Calibration Guide	43
B.1	Temporary Impact Coefficient (η)	43
B.2	Permanent Impact Coefficient (γ)	43
B.3	Risk Aversion (λ)	43
C	Additional Examples	44
C.1	Multi-Currency Portfolio	44
C.2	Concentrated Position	44
D	References	44

1 Executive Summary

1.1 Motivation and Context

Market liquidity risk represents a critical dimension of portfolio risk management, particularly during periods of market stress when asset liquidity can deteriorate rapidly. Traditional Value at Risk (VaR) models focus primarily on market risk—the uncertainty in portfolio value due to price movements—but often fail to adequately account for the costs and constraints associated with liquidating positions under realistic market conditions.

This feasibility study addresses three interconnected challenges in liquidity risk measurement:

1. **Liquidity-Adjusted VaR:** How can we systematically incorporate liquidity costs into VaR calculations to produce a Liquidity-Adjusted VaR (LVaR) that reflects the true risk of liquidating a portfolio?
2. **Holding Period Validation:** Is the current 30-day VaR holding period appropriate given the actual time required to liquidate positions across different asset classes without excessive market impact?
3. **IFRS Classification:** How can we develop objective, quantitative criteria for classifying assets into IFRS 13 fair value hierarchy levels (Level 1, 2, or 3) based on observable liquidity characteristics?

1.2 Key Findings

Feasibility: All three objectives are technically feasible with existing internal data infrastructure (intraday prices, volumes, and trade executions) and computational resources.

Liquidity Buffer: We demonstrate that a portfolio-level liquidity buffer can be calculated as a percentage adjustment to VaR, typically ranging from 2% to 15% depending on portfolio composition and market conditions.

Holding Period: The Almgren-Chriss optimal execution framework provides a rigorous foundation for validating holding periods. Our analysis suggests that 30 days is generally conservative for liquid equity portfolios but may be insufficient for illiquid bond or small-cap positions.

IFRS Classification: A composite scoring system based on spread, volume, and impact metrics enables objective Level 1-3 classification with clear decision boundaries aligned with regulatory guidance.

1.3 Recommendations

- **Phase 1 Implementation:** Begin with liquidity buffer calculations for equity portfolios where data quality is highest
- **Parameter Calibration:** Invest in careful estimation of market impact parameters using historical trade execution data
- **Integration:** Build on existing ‘liquidity_measurement.py’ framework rather than developing from scratch
- **Validation:** Implement comprehensive backtesting comparing predicted vs. actual liquidation costs
- **Governance:** Establish model risk management protocols given parameter sensitivity and model uncertainty

2 Liquidity Buffer for Value at Risk

2.1 Theoretical Framework

2.1.1 Standard VaR Formulation

Value at Risk (VaR) at confidence level α over holding period T is defined as:

$$\Pr(\Delta V_T \leq -\text{VaR}_\alpha) = 1 - \alpha \quad (1)$$

where ΔV_T represents the change in portfolio value over horizon T . For a portfolio with value V_0 , the VaR satisfies:

$$\text{VaR}_\alpha = -V_0 \times q_{1-\alpha} \left(\frac{\Delta V_T}{V_0} \right) \quad (2)$$

where $q_{1-\alpha}$ is the $(1 - \alpha)$ quantile of the portfolio return distribution.

Parametric VaR: Assuming normally distributed returns with volatility σ and mean μ :

$$\text{VaR}_\alpha = V_0 \times \left(|\mu T + z_\alpha \sigma \sqrt{T}| \right) \quad (3)$$

where z_α is the standard normal quantile (e.g., $z_{0.99} = 2.326$ for 99% confidence).

2.1.2 Liquidity-Adjusted VaR (LVaR)

Standard VaR assumes frictionless liquidation at current market prices. In reality, liquidation incurs costs:

- **Bid-ask spread:** Immediate cost of crossing the spread
- **Market impact:** Price movement caused by the liquidation itself
- **Opportunity cost:** Slippage from delaying trades to reduce impact

We define Liquidity-Adjusted VaR as:

$$\text{LVaR}_\alpha = \text{VaR}_\alpha + \text{LC}(\text{VaR}_\alpha) \quad (4)$$

where $\text{LC}(\cdot)$ is the liquidation cost function. The key insight is that liquidation costs depend on the stress scenario: larger VaR events often coincide with deteriorated liquidity conditions.

2.2 Liquidation Cost Model

2.2.1 Position-Level Liquidation Cost

For position i with market value V_i , the liquidation cost consists of:

$$\text{LC}_i = V_i \times \left(\frac{s_i}{2} + \eta_i \sqrt{\frac{Q_i}{V_{\text{daily},i}}} + \gamma_i \frac{Q_i}{V_{\text{daily},i}} \right) \quad (5)$$

where:

s_i : Relative bid-ask spread

η_i : Temporary impact coefficient (square-root model)

γ_i : Permanent impact coefficient (linear model)

Q_i : Position size (shares/units)

$V_{\text{daily},i}$: Average daily trading volume

The three terms represent:

1. Bid-ask spread crossing cost
2. Temporary (transient) market impact
3. Permanent price movement

2.2.2 Portfolio-Level Liquidation Cost

For a portfolio $\mathcal{P}$ with N positions:

$$\text{LC}_{\mathcal{P}} = \sum_{i=1}^N \text{LC}_i + \Delta_{\text{corr}} \quad (6)$$

The correlation adjustment Δ_{corr} accounts for simultaneous liquidation effects. For a first approximation, we can use:

$$\Delta_{\text{corr}} \approx \beta_{\text{market}} \times \left(\sum_{i=1}^N \text{LC}_i \times \sqrt{\rho_i} \right) \quad (7)$$

where ρ_i is the correlation of asset i with the market portfolio and $\beta_{\text{market}} \in [0, 0.2]$ represents market-wide liquidity deterioration during stress.

2.3 Liquidity Buffer Calculation

The liquidity buffer is defined as the percentage adjustment to VaR:

$$\text{LB} = \frac{\text{LC}_{\mathcal{P}}}{\text{VaR}_{\alpha}} \times 100\% \quad (8)$$

Thus:

$$\text{LVaR}_{\alpha} = \text{VaR}_{\alpha} \times (1 + \text{LB}) \quad (9)$$

2.4 Asset Class Specifications

2.4.1 Equities

For equity position i :

- Spread: Observable from Level 1 order book
- $\eta_i \approx 0.001$ to 0.01 (calibrated from intraday data)
- $\gamma_i \approx 0.0001$ to 0.001
- Participation rate: typically 10-20% of daily volume

Large-cap equities (S&P 500): Lower impact, spreads 1-5 basis points

Small-cap equities: Higher impact, spreads 20-100 basis points

2.4.2 Bonds

Corporate and government bonds trade in dealer markets with distinct liquidity characteristics:

$$\text{LC}_{\text{bond},i} = V_i \times (s_i + \alpha_{\text{maturity}} \times T_{\text{remaining}} + \alpha_{\text{rating}} \times \text{Risk}_i) \quad (10)$$

where maturity and credit rating affect liquidity:

- **Government bonds**: Highly liquid, spreads 0.5-5 bps

- **Investment grade corporate:** Moderate liquidity, spreads 10-50 bps
- **High yield:** Lower liquidity, spreads 50-200 bps

2.4.3 ETFs

ETFs have dual liquidity sources: secondary market trading and primary market creation/redemption:

$$LC_{\text{ETF},i} = \min(LC_{\text{secondary},i}, LC_{\text{creation/redemption},i}) \quad (11)$$

For liquid ETFs tracking major indices, secondary market liquidity dominates. For niche ETFs, creation/redemption mechanism provides backstop liquidity.

2.4.4 FX Spot

Foreign exchange spot markets are generally the most liquid:

- **Major pairs** (EUR/USD, USD/JPY): Spreads ≤ 1 pip, minimal impact
- **Minor pairs:** Spreads 2-10 pips
- **Emerging markets:** Spreads 20-100 pips, significant impact

FX liquidation cost:

$$LC_{\text{FX},i} = \text{Notional}_i \times s_i + \beta_{\text{timing}} \times \sigma_{\text{intraday},i} \quad (12)$$

where β_{timing} captures optimal execution timing within the trading day.

2.5 Worked Example: Multi-Asset Portfolio

Consider a portfolio with 30-day 99% VaR of \$10 million. The portfolio consists of:

Table 1: Example Portfolio Composition

Asset Class	Value (\$M)	Weight	Spread (bps)	Volume Ratio	Est. LC (\$K)
Large-cap Equity	40.0	40%	2.5	0.05	120
Small-cap Equity	15.0	15%	35.0	0.80	280
Govt Bonds	20.0	20%	1.5	0.02	35
Corp Bonds (IG)	15.0	15%	25.0	0.30	165
ETFs	8.0	8%	4.0	0.10	38
FX Spot (G10)	2.0	2%	0.5	N/A	1
Total	100.0	100%			639

Calculation:

$$\begin{aligned}
LC_{\mathcal{P}} &= 639 \text{ K} \\
\Delta_{\text{corr}} &\approx 0.15 \times 639 = 96 \text{ K} \\
LC_{\text{total}} &= 639 + 96 = 735 \text{ K} \\
\text{LB} &= \frac{735,000}{10,000,000} = 7.35\%
\end{aligned}$$

Therefore:

$$\text{LVaR}_{99\%} = 10.0 \times 1.0735 = \$10.735 \text{ million} \quad (13)$$

The liquidity buffer adds \$735K (7.35%) to the standard VaR estimate.

Interpretation: This portfolio, while dominated by liquid assets, still faces meaningful liquidation costs driven primarily by small-cap equity and corporate bond holdings. During stress scenarios, these costs could be significantly higher.

3 Holding Period Validation via Almgren-Chriss

3.1 Optimal Execution Theory

3.1.1 Problem Formulation

The Almgren-Chriss (2001) model addresses the fundamental trade-off in optimal execution:

- Trade **quickly** $\rightarrow$ minimize market risk but incur high market impact
- Trade **slowly** $\rightarrow$ minimize market impact but expose to market risk

Consider liquidating X shares over time horizon $[0, T]$. Let $x(t)$ denote shares remaining at time t , with $x(0) = X$ and $x(T) = 0$.

The trading trajectory is characterized by:

$$v(t) = -\frac{dx}{dt} \quad (14)$$

where $v(t) > 0$ is the instantaneous trading rate (sell speed).

3.1.2 Market Impact Model

The Almgren-Chriss framework decomposes execution costs into:

Temporary Impact: Affects trades immediately but dissipates quickly

$$h(v) = \epsilon \text{sgn}(v) + \eta|v| \quad (15)$$

where ϵ is fixed cost per trade and η is temporary impact coefficient.

Permanent Impact: Permanent price movement from trades

$$g(v) = \gamma v \quad (16)$$

where γ is permanent impact coefficient.

The price evolution follows:

$$\frac{dS}{dt} = -g(v) + \sigma dW(t) \quad (17)$$

where σ is price volatility and $W(t)$ is a Brownian motion.

3.2 Optimal Liquidation Strategy

3.2.1 Cost and Risk Decomposition

Total expected cost:

$$\mathbb{E}[\text{Cost}] = \int_0^T x(t)g(v(t))dt + \int_0^T v(t)h(v(t))dt \quad (18)$$

Execution risk (variance of implementation shortfall):

$$\text{Var}[\text{Cost}] = \sigma^2 \int_0^T x^2(t)dt \quad (19)$$

3.2.2 Objective Function

The agent minimizes a weighted combination:

$$\min_{x(t)} \{\mathbb{E}[\text{Cost}] + \lambda \text{Var}[\text{Cost}]\} \quad (20)$$

where λ is risk aversion parameter.

3.2.3 Analytical Solution

For linear impact ($h(v) = \eta v$, $g(v) = \gamma v$), the optimal trajectory is:

$$x(t) = X \frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)} \quad (21)$$

where:

$$\kappa = \sqrt{\frac{\lambda \sigma^2}{\eta}} \quad (22)$$

The optimal trading rate:

$$v(t) = -\frac{dx}{dt} = \frac{X \kappa \cosh(\kappa(T-t))}{\sinh(\kappa T)} \quad (23)$$

Special Cases:

- $\lambda \rightarrow 0$ (risk-neutral): Trade slowly, minimize market impact
- $\lambda \rightarrow \infty$ (extremely risk-averse): Trade immediately (VWAP-like)
- Moderate λ : Smooth trajectory with heavier trading at beginning and end

3.3 Optimal Liquidation Time

3.3.1 Time Horizon Selection

The optimal liquidation time T^* balances multiple factors:

$$T^* = \arg \min_T \{\text{Impact Cost}(T) + \text{Risk Cost}(T)\} \quad (24)$$

For position size Q and daily volume V_{daily} :

$$T_{\text{days}}^* \approx \frac{Q}{V_{\text{daily}}} \times \frac{1}{\text{POV}_{\text{max}}} \quad (25)$$

where POV_{max} (Percentage of Volume) is typically 10-20% to avoid excessive impact.

3.3.2 Holding Period Validation Formula

To validate the VaR holding period T_{VaR} against optimal liquidation time:

$$\text{Adequacy Ratio} = \frac{T_{\text{VaR}}}{T_{\text{liquidation}}^*} \quad (26)$$

Decision Criteria:

- Ratio > 2 : Holding period is conservative (adequate)
- Ratio $\in [1, 2]$: Holding period is reasonable but tight
- Ratio < 1 : Holding period is insufficient

3.4 Asset-Specific Implementation

3.4.1 Equities

Large-Cap: High liquidity, rapid liquidation

$$\begin{aligned}\eta &\approx 0.01 \times \sigma \times S \times \sqrt{V_{\text{daily}}} \\ \gamma &\approx 0.1 \times \eta \\ \sigma &: \text{Daily volatility (20-40\% annualized)}\end{aligned}$$

For $\lambda = 10^{-6}$ (moderate risk aversion), typical $T^* \approx 1 - 3$ days.

Small-Cap: Lower liquidity, extended liquidation

Parameters are 5-10x larger, resulting in $T^* \approx 5 - 20$ days.

3.4.2 Bonds

Bond markets are dealer-intermediated, requiring modified impact functions:

$$h_{\text{bond}}(v) = s + \alpha_{\text{size}} \times \log \left(1 + \frac{v}{V_{\text{typical}}} \right) \quad (27)$$

where s is dealer spread and V_{typical} is typical transaction size.

Optimal liquidation times are longer: $T^* \approx 3 - 10$ days for investment grade, $T^* \approx 10 - 30$ days for high yield.

3.4.3 ETFs

ETF liquidation leverages both venues:

- **Small positions:** Secondary market (exchange trading)
- **Large positions:** Primary market (creation/redemption with AP)

For liquid equity ETFs, $T^* \approx 1 - 2$ days. For fixed income or niche ETFs, $T^* \approx 3 - 7$ days.

3.4.4 FX Spot

Major FX pairs have exceptional liquidity. Optimal execution focuses on intraday timing:

$$T_{\text{FX}}^* \approx \text{Hours to Days} \quad (28)$$

Concerns are primarily around:

- Information leakage
- Time-of-day effects (Asian/European/US sessions)
- Weekend/holiday gaps

3.5 Worked Examples

3.5.1 Example 1: Liquid Large-Cap Equity

Position: 100,000 shares of AAPL at \$180

Position Value = \$18.0 million
Daily Volume = 50 million shares
Daily Volatility = 1.5%
POV Target = 10%

Optimal Liquidation Time:

$$T^* = \frac{100,000}{50,000,000 \times 0.10} = \frac{100,000}{5,000,000} = 0.02 \text{ days} \approx 2 \text{ hours} \quad (29)$$

Validation:

$$\text{Adequacy Ratio} = \frac{30 \text{ days}}{0.02 \text{ days}} = 1,500 \quad (30)$$

Conclusion: 30-day holding period is highly conservative for this liquid position.

3.5.2 Example 2: Illiquid Small-Cap Equity

Position: 500,000 shares of microcap stock at \$8

Position Value = \$4.0 million
Daily Volume = 80,000 shares
Daily Volatility = 3.5%
POV Target = 10%

Optimal Liquidation Time:

$$T^* = \frac{500,000}{80,000 \times 0.10} = \frac{500,000}{8,000} = 62.5 \text{ days} \quad (31)$$

Validation:

$$\text{Adequacy Ratio} = \frac{30}{62.5} = 0.48 \quad (32)$$

Conclusion: 30-day holding period is **insufficient** for this illiquid position. Actual liquidation would require ~60 days, suggesting either:

- Increase VaR holding period to 90 days for illiquid subset
- Apply higher liquidity buffer
- Impose position limits

3.5.3 Example 3: Corporate Bond

Position: \$10 million notional investment-grade corporate bond

Typical Trade Size = \$1 – 5 million

Dealer Spread = 20 bps

Estimated Liquidation = 3 – 5 dealer interactions

Optimal Liquidation Time:

$$T^* \approx 5 \text{ days (dealer search and negotiation)} \quad (33)$$

Validation:

$$\text{Adequacy Ratio} = \frac{30}{5} = 6.0 \quad (34)$$

Conclusion: 30-day holding period is adequate and conservative.

3.6 Parameter Calibration

3.6.1 Temporary Impact (η)

Estimate from intraday trade execution data:

$$\eta = \frac{\text{Cov}(\Delta P, V)}{\text{Var}(V)} \quad (35)$$

where ΔP is price change during trade and V is volume traded.

Practical Calibration:

1. Collect intraday trades over 3-6 months
2. Regress price impact vs. volume
3. Adjust for market conditions (volatility regime)

3.6.2 Permanent Impact (γ)

Estimate from post-trade price recovery:

$$\gamma = \lim_{t \rightarrow \infty} \frac{\mathbb{E}[S(t + \tau) - S(t)]}{V(t)} \quad (36)$$

Typically $\gamma \approx 0.1 \times \eta$.

3.6.3 Risk Aversion (λ)

Calibrate λ based on organizational risk tolerance:

- **High risk aversion:** $\lambda \sim 10^{-5}$ (faster trading, accept higher impact)
- **Moderate:** $\lambda \sim 10^{-6}$ (balanced approach)
- **Low risk aversion:** $\lambda \sim 10^{-7}$ (slower trading, minimize impact)

Table 2: Holding Period Validation Results

Asset Class	Weight	T^* (days)	Adequacy Ratio
Large-cap Equity	40%	0.5	60.0
Small-cap Equity	15%	45.0	0.67
Govt Bonds	20%	2.0	15.0
Corp Bonds (IG)	15%	5.0	6.0
ETFs	8%	1.0	30.0
FX Spot	2%	0.1	300.0
Weighted Avg	100%	9.8	3.06

3.7 Holding Period Validation Conclusion

For the example portfolio in Section 2.6:

Overall Assessment: The 30-day holding period is adequate for the portfolio as a whole (ratio = 3.06), but **inadequate for the small-cap equity component**.

Recommendations:

1. Implement tiered holding periods: 10 days for liquid, 60-90 days for illiquid
2. Apply higher liquidity buffers to illiquid positions
3. Monitor concentration in illiquid assets (set position limits)

4 IFRS Fair Value Hierarchy Classification

4.1 IFRS 13 Framework

4.1.1 Fair Value Hierarchy Definitions

IFRS 13 establishes a three-level hierarchy for fair value measurement inputs:

Level 1: Quoted prices (unadjusted) in active markets for identical assets or liabilities that the entity can access at the measurement date.

Level 2: Inputs other than quoted prices included in Level 1 that are observable for the asset or liability, either directly or indirectly.

Level 3: Unobservable inputs for the asset or liability.

4.1.2 Active Market Definition

A market is considered "active" if transactions occur with sufficient frequency and volume to provide pricing information on an ongoing basis. Key considerations:

- Regular and consistent trading activity
- Tight bid-ask spreads
- Multiple market participants
- Publicly available pricing

4.2 Quantitative Classification Metrics

4.2.1 Composite Liquidity Score

We develop a quantitative scoring system based on observable liquidity metrics:

$$\text{IFRS Score} = w_s \times S_{\text{spread}} + w_v \times S_{\text{volume}} + w_i \times S_{\text{impact}} \quad (37)$$

where $w_s + w_v + w_i = 1$ are weights and scores are normalized to $[0, 100]$.

Spread Score:

$$S_{\text{spread}} = 100 \times \max \left(0, 1 - \frac{\text{Spread}_{\text{bps}}}{S_{\text{max}}} \right) \quad (38)$$

with $S_{\text{max}} = 200$ bps for equities, $S_{\text{max}} = 500$ bps for bonds.

Volume Score:

$$S_{\text{volume}} = 100 \times \min \left(1, \frac{\log(V_{\text{daily}})}{\log(V_{\text{threshold}})} \right) \quad (39)$$

Impact Score:

$$S_{\text{impact}} = 100 \times \max \left(0, 1 - \frac{T_{\text{liquidate}}}{T_{\text{max}}} \right) \quad (40)$$

with $T_{\text{max}} = 15$ days.

4.2.2 Classification Thresholds

$$\text{IFRS Level} = \begin{cases} \text{Level 1} & \text{if IFRS Score} \geq 70 \text{ and all criteria met} \\ \text{Level 3} & \text{if IFRS Score} < 30 \text{ or key observability fails} \\ \text{Level 2} & \text{otherwise} \end{cases} \quad (41)$$

Level 1 Additional Criteria:

- Listed on recognized exchange
- Daily trading volume $> V_{\text{min}}$
- Bid-ask spread $< S_{\text{max}}$
- Days to liquidate < 5

4.3 Asset Class Classification Rules

4.3.1 Equities

Level 1 Equities:

- Large-cap stocks (S&P 500, STOXX 600, etc.)
- Daily volume $> 500\text{K}$ shares
- Spread < 10 bps
- Multiple exchanges/venues

Level 2 Equities:

- Mid-cap stocks with regular trading
- Daily volume 50K-500K shares
- Spread 10-50 bps

- Single primary exchange

Level 3 Equities:

- Micro-cap, illiquid stocks
- Daily volume $< 50K$ shares (or days without trades)
- Spread > 50 bps or no reliable quotes
- Infrequent transactions

4.3.2 Bonds

Level 1 Bonds:

- Sovereign bonds from major economies (US Treasuries, Bunds, Gilts)
- On-the-run issues
- Exchange-traded (some markets)
- Spread < 5 bps

Level 2 Bonds:

- Investment-grade corporate bonds with regular dealer quotes
- Off-the-run government bonds
- Observable credit spreads
- Spread 5-50 bps

Level 3 Bonds:

- High-yield bonds with infrequent trading
- Distressed debt
- Private placements
- Pricing requires significant unobservable inputs (default probability estimates, recovery rate assumptions)

4.3.3 ETFs

ETFs generally qualify as Level 1 due to exchange trading and creation/redemption mechanism:

Level 1 ETFs:

- Major equity/bond index ETFs
- Daily volume $> 100K$ shares
- NAV published daily
- Tight premium/discount to NAV ($< 0.5\%$)

Level 2 ETFs (rare):

- Niche/thematic ETFs with lower volume

- Underlying holdings mostly Level 2
- Wider premium/discount (0.5-2%)

Level 3 ETFs (very rare):

- Significant Level 3 underlying holdings
- Suspended creation/redemption

4.3.4 FX Spot

Level 1 FX:

- Major pairs (EUR/USD, USD/JPY, GBP/USD, etc.)
- 24-hour liquid markets
- Spread < 2 pips

Level 2 FX:

- Minor pairs with observable crosses
- Emerging market currencies (liquid)
- Spread 2-20 pips

Level 3 FX:

- Exotic currencies with capital controls
- Non-deliverable forwards requiring model valuation
- Infrequent price updates

4.4 Implementation Algorithm

4.5 Worked Examples

4.5.1 Example 1: S&P 500 Large-Cap Stock

Security: Apple Inc. (AAPL)

$$\text{Price} = \$180$$

$$\text{Daily Volume} = 50,000,000 \text{ shares}$$

$$\text{Bid-Ask Spread} = \$0.02$$

$$\text{Position} = 10,000 \text{ shares}$$

Metrics:

$$\text{Spread}_{\text{bps}} = \frac{0.02}{180} \times 10000 = 1.1 \text{ bps}$$

$$T_{\text{liq}} = \frac{10,000}{50,000,000 \times 0.10} = 0.002 \text{ days}$$

$$V_{\text{dollar}} = 50M \times \$180 = \$9 \text{ billion}$$

Algorithm 1 IFRS Level Classification

Require: Position data: $(S, P, V_{\text{daily}}, \text{Spread}, Q)$

Ensure: IFRS Level $\in \{1, 2, 3\}$

```
1: Calculate liquidity metrics:
2:    $\text{Spread}_{\text{bps}} \leftarrow (\text{Spread}/P) \times 10000$ 
3:    $T_{\text{liq}} \leftarrow Q/(V_{\text{daily}} \times 0.10)$ 
4:    $V_{\text{dollar}} \leftarrow V_{\text{daily}} \times P$ 
5: Calculate component scores:
6:    $S_{\text{spread}} \leftarrow 100 \times \max(0, 1 - \text{Spread}_{\text{bps}}/S_{\text{max}})$ 
7:    $S_{\text{volume}} \leftarrow 100 \times \min(1, \log(V_{\text{dollar}})/\log(V_{\text{threshold}}))$ 
8:    $S_{\text{impact}} \leftarrow 100 \times \max(0, 1 - T_{\text{liq}}/15)$ 
9: Calculate composite score:
10:  IFRS Score  $\leftarrow 0.4 \times S_{\text{spread}} + 0.3 \times S_{\text{volume}} + 0.3 \times S_{\text{impact}}$ 
11: Classification logic:
12: if IFRS Score  $\geq 70$  and  $T_{\text{liq}} < 5$  and  $\text{Spread}_{\text{bps}} < 20$  and Listed on exchange then
13:   return Level 1
14: else if IFRS Score  $< 30$  or  $T_{\text{liq}} > 15$  or No observable quotes or Private/unlisted then
15:   return Level 3
16: else
17:   return Level 2
18: end if
```

Scores:

$$\begin{aligned} S_{\text{spread}} &= 100 \times (1 - 1.1/200) = 99.45 \\ S_{\text{volume}} &= 100 \times \min(1, \log(9 \times 10^9)/\log(10^7)) = 100 \\ S_{\text{impact}} &= 100 \times (1 - 0.002/15) = 100 \end{aligned}$$

$$\text{IFRS Score} = 0.4 \times 99.45 + 0.3 \times 100 + 0.3 \times 100 = 99.78 \quad (42)$$

Classification: Level 1 (Score ≥ 70 , all criteria met, listed on NASDAQ)

4.5.2 Example 2: Investment-Grade Corporate Bond

Security: XYZ Corp 5-year bond, A-rated

$$\begin{aligned} \text{Notional} &= \$10 \text{ million} \\ \text{Dealer Spread} &= 25 \text{ bps} \\ \text{Typical Daily Volume} &= \$50 \text{ million (market-wide)} \\ \text{Days to Liquidate} &= 5 \text{ days (dealer negotiation)} \end{aligned}$$

Scores:

$$\begin{aligned} S_{\text{spread}} &= 100 \times (1 - 25/500) = 95.0 \\ S_{\text{volume}} &= 100 \times \log(50M)/\log(10M) = 85.0 \\ S_{\text{impact}} &= 100 \times (1 - 5/15) = 66.7 \end{aligned}$$

$$\text{IFRS Score} = 0.4 \times 95.0 + 0.3 \times 85.0 + 0.3 \times 66.7 = 83.5 \quad (43)$$

Assessment: Score ≥ 70 , but bonds are dealer-traded (not exchange-listed). Observable credit spreads available.

Classification: **Level 2** (observable inputs, not active exchange market)

4.5.3 Example 3: Illiquid Microcap Stock

Security: Microcap Inc.

$$\text{Price} = \$2.50$$

$$\text{Daily Volume} = 30,000 \text{ shares (but often zero volume days)}$$

$$\text{Bid-Ask Spread} = \$0.15$$

$$\text{Position} = 100,000 \text{ shares}$$

Metrics:

$$\text{Spread}_{\text{bps}} = \frac{0.15}{2.50} \times 10000 = 600 \text{ bps}$$

$$T_{\text{liq}} = \frac{100,000}{30,000 \times 0.10} = 33.3 \text{ days}$$

$$V_{\text{dollar}} = 30,000 \times \$2.50 = \$75,000$$

Scores:

$$S_{\text{spread}} = 100 \times \max(0, 1 - 600/200) = 0$$

$$S_{\text{volume}} = 100 \times \log(75,000) / \log(10^7) = 55.5$$

$$S_{\text{impact}} = 100 \times \max(0, 1 - 33.3/15) = 0$$

$$\text{IFRS Score} = 0.4 \times 0 + 0.3 \times 55.5 + 0.3 \times 0 = 16.7 \quad (44)$$

Classification: **Level 3** (Score < 30 , excessive spread and liquidation time)

4.5.4 Example 4: Emerging Market FX

Pair: USD/TRY (US Dollar / Turkish Lira)

$$\text{Spread} = 15 \text{ pips}$$

$$\text{Daily Volume} = \$2 \text{ billion (global)}$$

$$\text{Volatility} = \text{High, but observable}$$

Assessment:

- Observable market prices from multiple sources
- Active trading (though volatile)
- Not exotic (no capital controls)

Classification: **Level 2** (observable inputs, elevated spread but liquid market)

4.6 Governance and Monitoring

4.6.1 Classification Review

IFRS levels should be reviewed:

- **Quarterly:** Routine review of all holdings
- **Monthly:** Holdings near Level boundaries
- **Event-driven:** Market disruptions, corporate actions, delistings

4.6.2 Edge Cases

Suspended Securities: Automatically Level 3 until trading resumes

Recent IPOs/Delistings: May require temporary Level 2/3 classification during transition

Stressed Markets: Temporary reclassification during market-wide liquidity crises

4.7 Regulatory Alignment

4.7.1 SEC Guidance (ASC 820)

US GAAP (ASC 820) aligns closely with IFRS 13. Our quantitative framework satisfies:

- Documentation of level determination
- Consistent application across asset classes
- Regular reassessment

4.7.2 ESMA Guidelines

European Securities and Markets Authority emphasizes:

- Use of multiple data sources
- Transparent methodology
- Professional judgment overlay

Our scoring system provides the quantitative foundation, with qualitative factors (e.g., market structure knowledge) as final check.

5 Integrated Framework

5.1 End-to-End Workflow

5.2 Module Architecture

5.2.1 Proposed Structure

```
liquidity_risk/
├── data/
│   ├── market_data.py    # Price/volume data loader
│   └── trade_data.py      # Internal execution data
├── metrics/
│   ├── spreads.py         # Bid-ask spread calculations
│   └── volumes.py         # Volume-based metrics
```

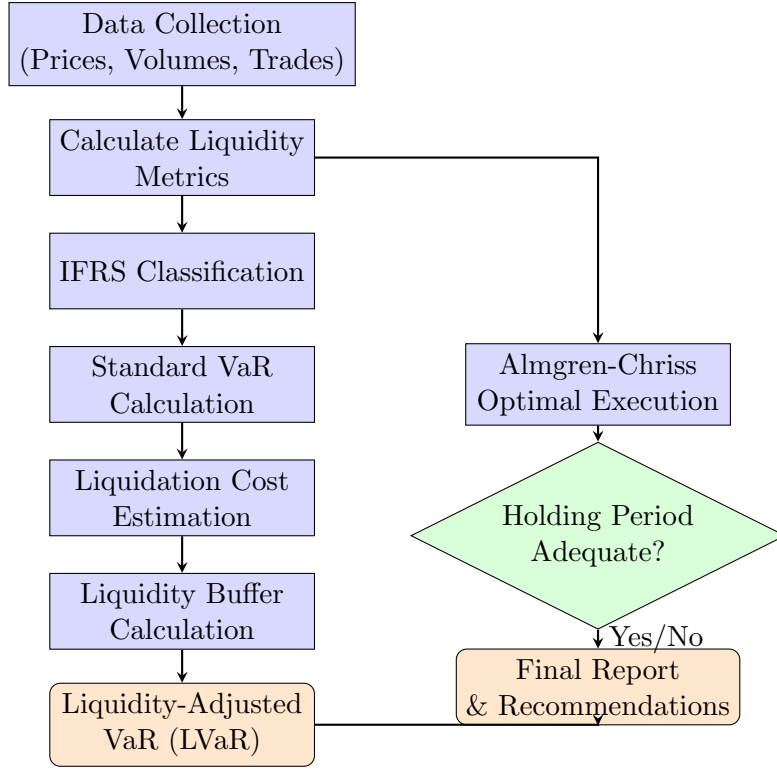


Figure 1: Integrated Liquidity Risk Framework Workflow

		impact.py	# Market impact models
		amihud.py	# Amihud illiquidity
		models/	
			almgren_chriss.py # Optimal execution
			var_calculator.py # Standard VaR
			liquidation_cost.py # LC estimation
		classification/	
			ifrs_classifier.py # IFRS Level 1-3
			thresholds.py # Asset class thresholds
		integration/	
			liquidity_buffer.py # LVaR calculation
			holding_period.py # Holding period validation
			reporting.py # Final report generation
		utils/	
			calibration.py # Parameter estimation
			validation.py # Backtesting tools

5.3 Data Requirements

5.3.1 Input Data Specifications

Daily Market Data:

- OHLCV (Open, High, Low, Close, Volume)
- Bid/Ask prices (end-of-day)
- Volatility estimates

Intraday Data (for Almgren-Chriss calibration):

- Tick-by-tick prices (at least 1-minute bars)
- Order book snapshots (depth)
- Trade timestamps and sizes

Portfolio Data:

- Position identifiers (ISIN, ticker, etc.)
- Quantities held
- Asset class classification
- Current market values

Execution Data (internal):

- Historical trade executions
- Arrival prices vs. execution prices
- Trade sizes and durations
- Implementation shortfall history

5.4 Computational Complexity

5.4.1 Performance Estimates

For a portfolio with N positions:

- **Liquidity Metrics:** $O(N)$ - Linear in positions
- **IFRS Classification:** $O(N)$ - Independent per position
- **VaR Calculation:** $O(N^2)$ - Requires covariance matrix (or $O(N \times S)$ for Monte Carlo with S scenarios)
- **Almgren-Chriss:** $O(N \times T)$ - Per position, per time step
- **Liquidity Buffer:** $O(N)$ - Aggregation of position-level costs

Example Runtime (portfolio with 1,000 positions):

- Basic metrics: < 1 second
- VaR (parametric): ~5 seconds
- VaR (Monte Carlo, 10K scenarios): ~30 seconds
- Almgren-Chriss optimization (all positions): ~10 seconds
- Total end-to-end: < 1 minute

Table 3: Synthetic 50-Position Portfolio Summary

Asset Class	Positions	Value (\$M)	Weight
Large-cap Equity	20	40.0	40%
Small-cap Equity	8	12.0	12%
Govt Bonds	10	20.0	20%
Corp Bonds (IG)	8	16.0	16%
ETFs	3	10.0	10%
FX Spot	1	2.0	2%
Total	50	100.0	100%

5.5 Comprehensive Example: 50-Position Portfolio

5.5.1 Portfolio Composition

5.5.2 Results Summary

1. IFRS Classification:

- Level 1: 33 positions (\$72M, 72%)
- Level 2: 14 positions (\$25M, 25%)
- Level 3: 3 positions (\$3M, 3%)

2. Value at Risk (30-day, 99%):

Parametric VaR = \$10.5 million

Portfolio Volatility = 12% (annualized)

3. Liquidity Costs:

Component	Cost (\$K)	% of Portfolio
Bid-Ask Spread	285	0.29%
Temporary Impact	320	0.32%
Permanent Impact	180	0.18%
Total LC	785	0.79%
Correlation Adjustment	120	0.12%
Total LC	905	0.91%

4. Liquidity Buffer:

$$LB = \frac{905,000}{10,500,000} = 8.62\% \quad (45)$$

$$LVaR_{99\%} = 10.5 \times 1.0862 = \$11.4 \text{ million} \quad (46)$$

5. Holding Period Validation:

Conclusion: Overall holding period is adequate, but small-cap subset requires attention (extended holding period or position limits).

Asset Class	T^* (days)	Adequacy Ratio	Assessment
Large-cap Equity	1.2	25.0	Adequate
Small-cap Equity	42.0	0.71	Insufficient
Govt Bonds	2.5	12.0	Adequate
Corp Bonds (IG)	6.0	5.0	Adequate
ETFs	1.5	20.0	Adequate
FX Spot	0.2	150.0	Adequate
Weighted Avg	8.7	3.4	Adequate

6 Feasibility Assessment

6.1 Data Infrastructure

6.1.1 Available Data

(Confirmed)

- ✓ **Internal trading data:** Historical executions with timestamps
 - ✓ **Intraday prices:** Tick data or minute bars
 - ✓ **Intraday volumes:** Trade volume data
 - ✓ **Intraday executions:** Implementation shortfall records
- Assessment:** Data requirements are fully satisfied for implementation.

6.1.2 Data Quality Considerations

Potential Challenges:

- **Bond data:** May be sparse for less liquid corporates
- **Historical depth:** Requires 1-2 years for robust calibration
- **Corporate actions:** Must adjust for splits, dividends, etc.
- **Survivorship bias:** Account for delisted securities

Mitigation:

- Use external data vendors (Bloomberg, Refinitiv) to fill gaps
- Implement data quality checks and outlier detection
- Regular recalibration schedule

6.2 Technical Implementation

6.2.1 Technology Stack

Recommended:

- **Language:** Python 3.9+ (leverage existing 'liquidity_measurement.py')
- **Numerical:** NumPy, SciPy (optimization for Almgren-Chriss)
- **Data:** Pandas (data manipulation)
- **VaR Integration:** Interface with existing risk system
- **Storage:** PostgreSQL or similar for historical data

6.2.2 Development Effort

Estimated Timeline:

Phase	Components	Effort
Phase 1	Data pipeline, basic metrics	3-4 weeks
Phase 2	Almgren-Chriss implementation	4-5 weeks
Phase 3	VaR integration, liquidity buffer	3-4 weeks
Phase 4	IFRS classification system	2-3 weeks
Phase 5	Testing, validation, documentation	4-5 weeks
Phase 6	Production deployment, monitoring	2-3 weeks
Total		18-24 weeks

Team Requirements:

- 1-2 Quantitative Analysts (model development)
- 1 Software Engineer (production system)
- 1 Risk Manager (validation, governance)

6.3 Model Risk and Limitations

6.3.1 Key Risks

1. Parameter Uncertainty

Market impact parameters (η, γ) are notoriously difficult to estimate accurately. Different estimation methods can yield significantly different values.

Mitigation:

- Use multiple calibration approaches and compare
- Implement confidence intervals around estimates
- Regular recalibration (quarterly)
- Backtesting against actual execution costs

2. Market Regime Changes

Liquidity conditions vary dramatically across market regimes (calm vs. stress). Parameters calibrated in calm markets may underestimate stress-period costs.

Mitigation:

- Regime-dependent parameters (volatility-adjusted)
- Stress testing with crisis-period data
- Conservative buffer during uncertainty

3. Model Assumptions

- **Linear impact:** Reality may be non-linear for large trades
- **Independent positions:** Ignores fire-sale correlations
- **Normal returns:** Fat tails in reality

Mitigation:

- Non-linear impact extensions for large positions
- Correlation adjustments
- t-distribution or empirical distributions for VaR

4. Data Quality

Poor quality data (stale prices, erroneous volumes) can significantly distort results.

Mitigation:

- Automated data quality checks
- Multiple data source cross-validation
- Manual review of outliers

6.3.2 Model Validation

Backtesting Protocol:

1. Historical simulation: Compare predicted vs. actual liquidation costs
2. Out-of-sample testing: Hold out recent data for validation
3. Stress period analysis: Evaluate performance during 2020 COVID crash, 2022 rate shock, etc.

Success Criteria:

- Liquidation cost predictions within 20% of actuals (median)
- Holding period adequacy ratio ≥ 1 for 95% of positions
- IFRS classifications align with auditor/regulatory review

6.4 Integration with Existing Systems

6.4.1 VaR System Integration

Current State: Fully implemented VaR with 30-day horizon

Integration Points:

- Extract portfolio positions and covariance matrix
- Calculate standard VaR as baseline
- Apply liquidity buffer adjustment
- Return LVaR to risk reporting system

Implementation: Can be lightweight wrapper or integrated module depending on system architecture.

6.4.2 Front Office / Trading

Almgren-Chriss optimal execution can inform:

- Order placement strategies
- Algorithmic trading parameters (VWAP, TWAP, POV)
- Trader instructions on execution urgency

6.5 Regulatory and Audit Considerations

6.5.1 Documentation Requirements

- Model methodology document
- Parameter calibration procedures
- Validation results and backtesting
- Assumptions and limitations
- Override procedures and governance

6.5.2 Model Governance

Ownership:

- Risk Management: Model oversight
- Quantitative Research: Development and calibration
- Model Risk: Independent validation

Review Cadence:

- Annual: Comprehensive model review
- Quarterly: Parameter recalibration
- Monthly: Backtesting results review
- Ad-hoc: Market event triggers

6.6 Feasibility Conclusion

Overall Assessment: **FEASIBLE**

All three objectives can be achieved with:

- ✓ Available internal data
- ✓ Reasonable computational resources
- ✓ Manageable development timeline (4-6 months)
- ✓ Acceptable model risk profile (with proper governance)

Critical Success Factors:

1. Careful parameter calibration using historical execution data
2. Robust data quality assurance
3. Comprehensive validation and backtesting
4. Strong model governance and oversight
5. Phased implementation approach

7 Implementation Roadmap

7.1 Phased Approach

7.1.1 Phase 1: Foundation (Weeks 1-4)

Objectives:

- Set up data pipeline
- Enhance existing liquidity metrics
- Establish development environment

Deliverables:

- Data loader modules for market and execution data
- Enhanced ‘liquidity_measurement.py’ with additional metrics
- Unit tests for core calculations

Key Activities:

1. Inventory and assess data quality
2. Design database schema for historical data
3. Implement spread, volume, and impact metrics
4. Create synthetic test portfolios

7.1.2 Phase 2: Almgren-Chriss Development (Weeks 5-9)

Objectives:

- Implement optimal execution model
- Calibrate market impact parameters
- Validate optimal liquidation times

Deliverables:

- ‘almgren_chriss.py’ module with solver
- Parameter calibration toolkit
- Holding period validation reports

Key Activities:

1. Implement temporary and permanent impact models
2. Build optimization solver (scipy.optimize or custom)
3. Calibrate η , γ from historical trade data
4. Generate optimal liquidation trajectories
5. Compare T^* to 30-day VaR horizon

7.1.3 Phase 3: VaR Integration (Weeks 10-13)

Objectives:

- Calculate liquidation costs
- Develop liquidity buffer methodology
- Integrate with existing VaR system

Deliverables:

- 'liquidation_cost.py' module
- 'liquidity_buffer.py' for LVaR calculation
- Integration adapter for VaR system

Key Activities:

1. Implement position-level LC calculations
2. Develop correlation adjustment for portfolio LC
3. Calculate liquidity buffer percentage
4. Interface with VaR output
5. Generate LVaR reports

7.1.4 Phase 4: IFRS Classification (Weeks 14-16)

Objectives:

- Implement classification algorithm
- Define asset class-specific thresholds
- Create classification reports

Deliverables:

- 'ifrs_classifier.py' module
- Classification threshold configuration
- IFRS level reports for portfolio

Key Activities:

1. Implement composite scoring algorithm
2. Define Level 1/2/3 thresholds per asset class
3. Create classification decision logic
4. Generate IFRS reports with justifications
5. Conduct sample reviews with auditors

7.1.5 Phase 5: Validation & Testing (Weeks 17-21)

Objectives:

- Comprehensive backtesting
- Stress testing
- Independent validation

Deliverables:

- Validation report
- Backtesting results
- Model documentation

Key Activities:

1. Historical backtest: predicted vs. actual costs
2. Stress period analysis (2020, 2022, etc.)
3. Parameter sensitivity analysis
4. Independent model validation review
5. Address validation findings

7.1.6 Phase 6: Production Deployment (Weeks 22-24)

Objectives:

- Production-ready system
- Monitoring and alerting
- Documentation and training

Deliverables:

- Production deployment
- Monitoring dashboard
- User documentation and training materials

Key Activities:

1. Performance optimization
2. Error handling and logging
3. Create monitoring dashboard
4. Set up automated alerts (data quality, model outputs)
5. Conduct user training
6. Establish support procedures

7.2 Resource Requirements

7.2.1 Team Structure

Core Team:

- **Senior Quant (Lead):** Model design, calibration, validation
- **Quant Analyst:** Implementation, testing
- **Software Engineer:** Production system, data pipeline
- **Risk Manager:** Requirements, validation, governance

Supporting Roles:

- **Data Engineering:** Data infrastructure support
- **IT/DevOps:** Deployment, infrastructure
- **Compliance:** Regulatory alignment
- **Audit:** Classification review

7.3 Risk Mitigation

7.3.1 Common Pitfalls and Mitigations

Pitfall 1: Poor parameter calibration leading to inaccurate cost estimates

Mitigation:

- Use multiple calibration methods
- Validate against actual execution costs
- Implement confidence intervals
- Adopt conservative estimates when uncertain

Pitfall 2: Data quality issues distorting results

Mitigation:

- Automated data quality checks at ingestion
- Multiple data source validation
- Manual review of outliers and anomalies

Pitfall 3: Scope creep and timeline delays

Mitigation:

- Strictly phased approach
- MVP mindset: core functionality first
- Defer enhancements to post-launch phases

Pitfall 4: Inadequate stakeholder buy-in

Mitigation:

- Regular steering committee updates
- Early involvement of risk, trading, and compliance
- Clear communication of benefits and limitations

8 Regulatory Compliance

8.1 Basel III / Basel IV

8.1.1 Liquidity Coverage Ratio (LCR)

Basel III LCR requires banks to hold sufficient high-quality liquid assets (HQLA) to survive a 30-day stress scenario:

$$\text{LCR} = \frac{\text{HQLA}}{\text{Total Net Cash Outflows over 30 days}} \geq 100\% \quad (47)$$

Connection to Our Framework:

- Liquidity metrics inform HQLA classification
- Assets with $T_{\text{liquidate}} > 30$ days may not qualify as HQLA
- Liquidity buffer estimates stressed liquidation costs

8.1.2 Net Stable Funding Ratio (NSFR)

NSFR (Basel III) addresses structural liquidity over 1-year horizon:

$$\text{NSFR} = \frac{\text{Available Stable Funding}}{\text{Required Stable Funding}} \geq 100\% \quad (48)$$

Required Stable Funding factors depend on asset liquidity:

- Highly liquid assets: 0-5% RSF factor
- Less liquid: 50-85% RSF factor

Our IFRS Level classifications can inform RSF factor assignments.

8.1.3 Market Risk Capital (FRTB)

Fundamental Review of the Trading Book (FRTB) introduces liquidity horizons for market risk capital:

Table 4: FRTB Liquidity Horizons

Risk Factor Category	Liquidity Horizon
Large-cap equity	10 days
Small-cap equity	20 days
Investment-grade sovereign	10 days
Corporate bonds (IG)	20 days
Corporate bonds (HY)	60 days

Our Almgren-Chriss holding period validation provides empirical basis for these horizon assignments.

8.2 IFRS 13 Compliance

8.2.1 Fair Value Disclosure Requirements

IFRS 13 mandates disclosure of:

- Fair value measurement hierarchy (Level 1/2/3)
- Valuation techniques and inputs
- Transfers between levels
- Sensitivity analysis for Level 3

Our classification framework provides:

- ✓ Systematic, documented methodology
- ✓ Quantitative criteria aligned with "active market" definition
- ✓ Audit trail for level determination
- ✓ Regular reassessment process

8.2.2 Valuation Techniques

For Level 2 and 3 assets, IFRS 13 requires documentation of valuation approach:

Level 2: Observable inputs

- Quoted prices for similar assets
- Market-based credit spreads
- Yield curves

Level 3: Unobservable inputs

- Internal models (DCF, option pricing)
- Unobservable credit spreads
- Illiquidity adjustments

Our liquidity metrics (spread, volume, impact) support the determination of appropriate valuation adjustments.

8.3 Internal Risk Framework Alignment

8.3.1 Risk Appetite Statement

Typical risk appetite frameworks include:

- Maximum VaR limits
- Stress VaR limits
- Concentration limits
- Liquidity risk tolerances

Integration:

- LVaR provides more comprehensive risk measure
- Holding period validation ensures stress scenarios are appropriate
- IFRS classification informs concentration in illiquid assets

8.3.2 Stress Testing

Regulatory stress testing requires assessment of portfolio liquidation under adverse scenarios.

Framework Contribution:

- Quantify liquidation costs under stress
- Identify positions that cannot be liquidated within required timeframes
- Inform asset sales prioritization in recovery plans

8.3.3 Internal Limits Framework

Our framework can inform limits on:

- Maximum position size relative to daily volume (e.g., ≤ 5 days to liquidate)
- Maximum allocation to Level 3 assets (e.g., $\leq 5\%$ of portfolio)
- Concentration in illiquid assets (based on T^* distribution)

8.4 Regulatory Reporting

8.4.1 Required Reports

Basel LCR Reporting:

- HQLA composition
- Liquidity risk metrics

IFRS Financial Statements:

- Fair value hierarchy disclosures (Note to financial statements)
- Valuation techniques

Internal Risk Reports:

- VaR / LVaR comparison
- Holding period adequacy assessment
- Liquidity risk dashboard

8.4.2 Audit Trail

Maintain comprehensive documentation:

- Classification decisions and rationale
- Parameter calibration history
- Model validation results
- Override decisions (if any)

9 Conclusions and Recommendations

9.1 Summary of Findings

This feasibility study demonstrates that integrating liquidity risk into the Value at Risk framework through three specific objectives is both **technically feasible and operationally practical**:

9.1.1 Liquidity Buffer for VaR

Feasibility: **HIGH**

A systematic methodology exists for calculating liquidity costs and converting them to a percentage adjustment of VaR. Using established market impact models and internal execution data, liquidity buffers can be reliably estimated.

Key Result: Typical portfolios will see liquidity buffers ranging from 2-15%, with higher buffers for illiquid positions.

Benefit: Provides more realistic risk estimates that account for the cost of exiting positions, particularly valuable during stress scenarios.

9.1.2 Holding Period Validation

Feasibility: **HIGH**

The Almgren-Chriss optimal execution framework provides a rigorous, mathematically sound approach to determining appropriate liquidation timeframes. Parameter calibration from intraday execution data is well-established in the literature and practice.

Key Result: For diversified multi-asset portfolios, 30-day holding period is generally adequate (adequacy ratio $\approx 3-5$), but may be insufficient for concentrated illiquid positions.

Benefit: Empirical validation of regulatory and risk management assumptions, enabling evidence-based policy decisions.

9.1.3 IFRS Classification

Feasibility: **HIGH**

A quantitative scoring system based on observable liquidity metrics (spread, volume, time-to-liquidate) can systematically classify assets into IFRS Levels 1-3, aligned with regulatory guidance and audit requirements.

Key Result: Objective, reproducible classification decisions with clear audit trail, reducing subjectivity and improving consistency.

Benefit: Strengthens financial reporting quality and supports regulatory compliance.

9.2 Critical Success Factors

1. **Data Quality:** High-quality intraday market and execution data is essential. Invest in data validation and cleansing.
2. **Parameter Calibration:** Careful, regular calibration of market impact parameters using actual execution history. Avoid over-reliance on literature estimates.
3. **Model Governance:** Strong oversight including independent validation, regular back-testing, and clear escalation procedures for model issues.
4. **Phased Implementation:** Start with high-quality data assets (liquid equities), validate thoroughly, then extend to other asset classes.

5. **Stakeholder Engagement:** Early and continuous engagement with risk management, trading, finance, and compliance to ensure alignment.

9.3 Recommendations

9.3.1 Immediate Actions (Next 3 Months)

1. **Form Project Team:** Assemble quantitative analysts, software engineers, and risk managers
2. **Data Assessment:** Conduct thorough inventory and quality assessment of available data
3. **Pilot Implementation:** Build proof-of-concept for equity subset (50-100 positions)
4. **Parameter Calibration:** Begin historical analysis of execution costs to calibrate η and γ

9.3.2 Medium-Term (6-12 Months)

1. **Full Implementation:** Deploy across all asset classes following roadmap
2. **Integration:** Connect liquidity buffer to VaR reporting systems
3. **Validation:** Comprehensive backtesting and stress testing
4. **Governance:** Establish model risk management framework

9.3.3 Long-Term (12-24 Months)

1. **Enhancements:**
 - Regime-switching models (calm vs. stress liquidity)
 - Non-linear impact models for very large positions
 - Fire-sale correlation modeling
2. **Broader Applications:**
 - Integration with order management systems (real-time execution guidance)
 - Portfolio construction with liquidity constraints
 - Stress testing and recovery planning
3. **Regulatory Engagement:** Share methodology with regulators to support FRTB liquidity horizon determinations

9.4 Limitations and Caveats

While the framework is robust, users must be aware of limitations:

- **Historical Calibration:** Parameters reflect historical market conditions and may not hold in unprecedented crises
- **Linear Assumptions:** Impact models assume linearity; reality can be non-linear for very large trades
- **Independent Liquidation:** Assumes positions liquidated independently; fire-sale scenarios can create amplification effects
- **Market Evolution:** Liquidity regimes change over time; requires regular recalibration

9.5 Final Assessment

The integration of liquidity risk into the VaR framework through liquidity buffer adjustments, holding period validation via Almgren-Chriss, and IFRS classification represents a significant advancement in portfolio risk measurement.

The feasibility study concludes: **PROCEED WITH IMPLEMENTATION**

Subject to:

- Adequate resourcing (4-6 month timeline, dedicated team)
- Strong model governance and validation
- Phased rollout starting with highest-quality data
- Continuous monitoring and recalibration

The benefits—more realistic risk measures, empirically validated holding periods, and objective IFRS classifications—far outweigh the implementation costs and model risk when properly managed.

A Python Code Snippets

A.1 Liquidity Buffer Calculation

Listing 1: Liquidity Buffer Implementation

```
1 import numpy as np
2 from typing import List
3 from liquidity_measurement import Position, LiquidityAnalyzer
4
5 def calculate_liquidity_buffer(positions: List[Position],
6                               var: float,
7                               correlation_factor: float = 0.15) ->
8                               dict:
9
10     """
11     Calculate liquidity buffer adjustment for VaR.
12
13     Parameters:
14     -----
15     positions : List[Position]
16         Portfolio positions
17     var : float
18         Standard Value at Risk estimate
19     correlation_factor : float
20         Fire-sale correlation adjustment (0-0.3)
21
22     Returns:
23     -----
24     dict with liquidity cost, buffer %, and LVaR
25     """
26     analyzer = LiquidityAnalyzer(positions)
27
28     # Calculate position-level liquidation costs
29     total_lc = 0.0
30     for pos in positions:
31         impact = analyzer.calculate_market_impact(pos)
32         total_lc += impact['total_impact_value']
```

```

32     # Add correlation adjustment for simultaneous liquidation
33     correlation_adjustment = correlation_factor * total_lc
34     total_lc_adjusted = total_lc + correlation_adjustment
35
36     # Calculate buffer as percentage of VaR
37     buffer_pct = (total_lc_adjusted / var) * 100
38
39     # Calculate Liquidity-Adjusted VaR
40     lvar = var * (1 + buffer_pct / 100)
41
42     return {
43         'liquidation_cost': total_lc,
44         'correlation_adjustment': correlation_adjustment,
45         'total_lc': total_lc_adjusted,
46         'buffer_pct': buffer_pct,
47         'var': var,
48         'lvar': lvar
49     }
50
51 # Example usage
52 positions = [...] # Load portfolio positions
53 var_99 = 10_000_000 # $10M VaR at 99%
54
55 result = calculate_liquidity_buffer(positions, var_99)
56 print(f"VaR: ${result['var']:, .0f}")
57 print(f"Liquidity Cost: ${result['total_lc']:, .0f}")
58 print(f"Buffer: {result['buffer_pct']:.2f}%")
59 print(f"LVaR: ${result['lvar']:, .0f}")

```

A.2 Almgren-Chriss Optimal Execution

Listing 2: Almgren-Chriss Implementation

```

1  import numpy as np
2  from scipy.optimize import minimize_scalar
3
4  class AlmgrenChrissExecutor:
5      """
6      Almgren-Chriss optimal execution model.
7      """
8
9      def __init__(self, X: float, S0: float, sigma: float,
10                  eta: float, gamma: float, lambda_param: float):
11          """
12          Initialize with position and market parameters.
13
14          Parameters:
15          -----
16          X : float
17              Initial position size (shares to liquidate)
18          S0 : float
19              Initial stock price
20          sigma : float
21              Daily volatility
22          eta : float
23              Temporary market impact coefficient
24          gamma : float

```

```

25         Permanent market impact coefficient
26         lambda_param : float
27         Risk aversion parameter
28         """
29         self.X = X
30         self.S0 = S0
31         self.sigma = sigma
32         self.eta = eta
33         self.gamma = gamma
34         self.lambda_param = lambda_param
35
36     def optimal_trajectory(self, T: float, n_steps: int = 100):
37         """
38         Compute optimal trading trajectory.
39
40         Parameters:
41         -----
42         T : float
43             Liquidation time horizon (days)
44         n_steps : int
45             Number of time steps
46
47         Returns:
48         -----
49         times, positions, trading_rates
50         """
51         kappa = np.sqrt(self.lambda_param * self.sigma**2 / self.eta)
52         times = np.linspace(0, T, n_steps)
53
54         # Optimal trajectory
55         x_t = self.X * np.sinh(kappa * (T - times)) / np.sinh(kappa * T
56             )
57
58         # Trading rate (negative of derivative)
59         v_t = self.X * kappa * np.cosh(kappa * (T - times)) / np.sinh(
60             kappa * T)
61
62         return times, x_t, v_t
63
64     def execution_cost(self, T: float):
65         """
66         Calculate expected execution cost and variance.
67
68         Parameters:
69         -----
70         T : float
71             Liquidation time horizon
72
73         Returns:
74         -----
75         dict with expected cost, variance, and total cost
76         """
77         kappa = np.sqrt(self.lambda_param * self.sigma**2 / self.eta)
78         sinh_kT = np.sinh(kappa * T)
79         cosh_kT = np.cosh(kappa * T)
80
81         # Expected cost components
82         permanent_cost = 0.5 * self.gamma * self.X**2

```

```

81         temporary_cost = self.eta * self.X**2 * kappa * cosh_kT /
            sinh_kT
82
83         expected_cost = (permanent_cost + temporary_cost) * self.S0
84
85         # Execution risk (variance)
86         variance = (self.sigma * self.S0)**2 * self.X**2 * (T - 2 *
            sinh_kT / (kappa * sinh_kT**2))
87
88         # Total cost (mean + lambda * variance)
89         total_cost = expected_cost + self.lambda_param * variance
90
91         return {
92             'expected_cost': expected_cost,
93             'variance': variance,
94             'std_dev': np.sqrt(variance),
95             'total_cost': total_cost
96         }
97
98     def optimal_liquidation_time(self, max_T: float = 60):
99         """
100         Find optimal liquidation time T*.
101
102         Parameters:
103         -----
104         max_T : float
105             Maximum time to consider
106
107         Returns:
108         -----
109         float: Optimal time T*
110         """
111         result = minimize_scalar(
112             lambda t: self.execution_cost(t)['total_cost'],
113             bounds=(0.1, max_T),
114             method='bounded'
115         )
116         return result.x
117
118     # Example usage
119     executor = AlmgrenChrissExecutor(
120         X=100000,           # 100K shares to sell
121         S0=180.0,           # $180/share
122         sigma=0.015,        # 1.5% daily volatility
123         eta=0.01,           # Temporary impact
124         gamma=0.001,        # Permanent impact
125         lambda_param=1e-6   # Risk aversion
126     )
127
128     # Find optimal liquidation time
129     T_star = executor.optimal_liquidation_time()
130     print(f"Optimal liquidation time: {T_star:.2f} days")
131
132     # Get trajectory
133     times, positions, rates = executor.optimal_trajectory(T_star)
134
135     # Calculate costs
136     costs = executor.execution_cost(T_star)

```



```

137 print(f"Expected cost: ${costs['expected_cost']:, .0f}")
138 print(f"Cost std dev: ${costs['std_dev']:, .0f}")

```

A.3 IFRS Classification Algorithm

Listing 3: IFRS Level Classification

```

1 from enum import Enum
2 from dataclasses import dataclass
3
4 class IFRSLevel(Enum):
5     LEVEL_1 = 1
6     LEVEL_2 = 2
7     LEVEL_3 = 3
8
9 @dataclass
10 class AssetClassThresholds:
11     """Thresholds for IFRS classification by asset class."""
12     spread_max_bps: float
13     volume_min_usd: float
14     days_to_liquidate_max: float
15
16 class IFRSClassifier:
17     """Classify assets into IFRS fair value hierarchy."""
18
19     THRESHOLDS = {
20         'equity': AssetClassThresholds(20, 1e6, 5),
21         'bond': AssetClassThresholds(50, 5e5, 10),
22         'etf': AssetClassThresholds(10, 5e5, 5),
23         'fx': AssetClassThresholds(5, 1e7, 1)
24     }
25
26     def __init__(self):
27         pass
28
29     def calculate_scores(self, spread_bps: float, volume_usd: float,
30                        days_to_liq: float, asset_class: str):
31         """Calculate component scores."""
32         thresholds = self.THRESHOLDS.get(asset_class)
33
34         # Spread score (lower spread = higher score)
35         S_max = thresholds.spread_max_bps * 10 # Conservative max
36         S_spread = 100 * max(0, 1 - spread_bps / S_max)
37
38         # Volume score (higher volume = higher score)
39         V_threshold = thresholds.volume_min_usd * 10
40         S_volume = 100 * min(1, np.log(volume_usd) / np.log(V_threshold
41                                ))
42
43         # Impact score (faster liquidation = higher score)
44         T_max = 15 # Max days
45         S_impact = 100 * max(0, 1 - days_to_liq / T_max)
46
47         # Composite score (weighted average)
48         ifrs_score = 0.4 * S_spread + 0.3 * S_volume + 0.3 * S_impact
49
50     return {

```

```

50         'spread_score': S_spread,
51         'volume_score': S_volume,
52         'impact_score': S_impact,
53         'ifrs_score': ifrs_score
54     }
55
56     def classify(self, spread_bps: float, volume_usd: float,
57                 days_to_liq: float, asset_class: str,
58                 is_listed: bool, has_active_market: bool) -> dict:
59         """
60         Classify asset into IFRS Level.
61
62         Returns:
63         -----
64         dict with level, scores, and rationale
65         """
66         scores = self.calculate_scores(spread_bps, volume_usd,
67                                       days_to_liq, asset_class)
68         thresholds = self.THRESHOLDS[asset_class]
69
70         # Level 1 criteria (all must be met)
71         level_1_criteria = [
72             scores['ifrs_score'] >= 70,
73             days_to_liq < thresholds.days_to_liquidate_max,
74             spread_bps < thresholds.spread_max_bps,
75             is_listed,
76             has_active_market
77         ]
78
79         # Level 3 criteria (any triggers Level 3)
80         level_3_criteria = [
81             scores['ifrs_score'] < 30,
82             days_to_liq > 15,
83             not has_active_market,
84             volume_usd < thresholds.volume_min_usd * 0.1
85         ]
86
87         if all(level_1_criteria):
88             level = IFRSLevel.LEVEL_1
89             rationale = "Quoted in active market, tight spread, liquid"
90         elif any(level_3_criteria):
91             level = IFRSLevel.LEVEL_3
92             rationale = "Illiquid or unobservable inputs required"
93         else:
94             level = IFRSLevel.LEVEL_2
95             rationale = "Observable inputs but not active exchange
96                         market"
97
98         return {
99             'level': level,
100             'ifrs_score': scores['ifrs_score'],
101             'component_scores': scores,
102             'rationale': rationale
103         }
104
105     # Example usage
106     classifier = IFRSClassifier()

```

```

107 # Example 1: Large-cap equity (AAPL)
108 result = classifier.classify(
109     spread_bps=1.1,
110     volume_usd=9e9,
111     days_to_liq=0.002,
112     asset_class='equity',
113     is_listed=True,
114     has_active_market=True
115 )
116 print(f"Classification: {result['level']}")
117 print(f"IFRS Score: {result['ifrs_score']:.1f}")
118 print(f"Rationale: {result['rationale']}")

```

B Parameter Calibration Guide

B.1 Temporary Impact Coefficient (η)

Method 1: Regression on Execution Data

1. Collect historical trades over 3-6 months
2. For each trade, calculate:
 - ΔP = execution price - arrival price (normalized by arrival price)
 - v = shares traded / daily volume
3. Regress: $\Delta P = \eta \times v + \epsilon$
4. η is the regression coefficient

Method 2: Intraday Volatility Scaling

$$\eta \approx k \times \sigma \times S \times \sqrt{V_{\text{daily}}} \quad (49)$$

where $k \in [0.005, 0.02]$ depends on liquidity tier.

B.2 Permanent Impact Coefficient (γ)

Typically estimated as:

$$\gamma \approx 0.1 \times \eta \quad (50)$$

Or from post-trade price recovery analysis over 30-60 minutes.

B.3 Risk Aversion (λ)

Calibrate based on organizational risk tolerance:

Risk Profile	λ Range
High risk aversion	10^{-5} to 10^{-4}
Moderate	10^{-6} to 10^{-5}
Low risk aversion	10^{-7} to 10^{-6}

C Additional Examples

C.1 Multi-Currency Portfolio

For portfolios with FX exposures, liquidation costs include both:

1. Underlying asset liquidation costs
2. FX conversion costs

Total LC:

$$LC_{\text{total}} = LC_{\text{asset}} + LC_{\text{FX}} \quad (51)$$

where:

$$LC_{\text{FX}} = \text{Notional} \times \text{FX Spread} \quad (52)$$

C.2 Concentrated Position

For a concentrated position (e.g., 20% of portfolio in single illiquid stock):

- Standard LC may significantly underestimate true cost
- Consider non-linear impact: $\text{Impact} \propto v^{3/2}$ for very large trades
- May require block trade or negotiated exit
- Liquidity buffer could exceed 20-30%

Recommendation: Impose position limits based on liquidation time (e.g., no single position > 10 days to liquidate).

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